Homework 9

3.1.2 Consider the function $g = -\chi_{[a,\frac{a+b}{2}]} + \chi_{[\frac{a+b}{2},b]}$. Show that for all $x \notin [a,b]$ we have

$$0 < H(g)(x) < \frac{1}{4\pi}\frac{(b-a)^2}{(x-a)(x-b)}.$$

Moreover, obtain that $H(g)(x)$ is proportional to $|x|^{-2}$ as $|x| \to \infty$.

Proof.

$$H(g)(x) = \text{P.V.}\frac{1}{\pi}\int_{\mathbb{R}} \frac{g(x-t)}{t}dt = \text{P.V.}\frac{1}{\pi}\int_{\mathbb{R}} \frac{-\chi_{[a,\frac{a+b}{2}]}(x-t) + \chi_{[\frac{a+b}{2},b]}(x-t)}{t}dt.$$

We wish to split this integral at the discontinuities. $a < x-t < \frac{a+b}{2} \Rightarrow x - \frac{a+b}{2} < t < x-a$. And $\frac{a+b}{2} < x-t < b \Rightarrow x-b < t < x - \frac{a+b}{2}$. This gives us the new bounds at which we can split the integral. Show

$$\begin{aligned}
\text{P.V.}\frac{1}{\pi}\int_{\mathbb{R}} \frac{-\chi_{[a,\frac{a+b}{2}]}(x-t) + \chi_{[\frac{a+b}{2},b]}(x-t)}{t}dt &= \text{P.V.}\frac{1}{\pi}\int_{x-b}^{x-\frac{a+b}{2}} \frac{dt}{t} - \text{P.V.}\frac{1}{\pi}\int_{x-\frac{a+b}{2}}^{x-a} \frac{dt}{t} \\
&= H(\chi_{[\frac{a+b}{2},b]})(x) - H(\chi_{[a,\frac{a+b}{2}]})(x) \\
&= \frac{1}{\pi}\log\frac{|x-\frac{a+b}{2}|}{|x-b|} + \frac{1}{\pi}\log\frac{|x-\frac{a+b}{2}|}{|x-a|} \\
&= \frac{1}{\pi}\log\frac{(x-\frac{a+b}{2})^2}{|x-b||x-a|}.
\end{aligned}$$

We intend to use the identity $\log(1+y) < y$ for $y > 0$, but first we must show that $\frac{(x-\frac{a+b}{2})^2}{|x-a||x-b|} > 1$.

$$\begin{aligned}
\frac{(x-\frac{a+b}{2})^2}{(x-a)(x-b)} - 1 &= \frac{(x-\frac{a+b}{2})^2 - (x-b)(x-a)}{(x-b)(x-a)} \\
&= \frac{x^2 - (a+b)x + \frac{(a+b)^2}{4} - x^2 + (a+b)x - ab}{(x-b)(x-a)} \\
&= \frac{\frac{(a+b)^2-4ab}{4}}{(x-b)(x-a)} = \frac{a^2-2ab^2}{4(x-b)(x-a)} = \frac{(b-a)^2}{4(x-a)(x-b)} > 0.
\end{aligned}$$

Hence we can now apply the logarithm identity.

$$\frac{1}{\pi}\log(\frac{(x-\frac{a+b}{2})^2}{(x-b)(x-a)}) < \frac{1}{4\pi}\frac{(b-a)^2}{(x-a)(x-b)}.$$

We now show that $H(g)(x)$ is proportional to $|x|^{-2}$ as $|x| \to \infty$.

$$\lim_{|x|\to\infty} \frac{\log \frac{(x-\frac{a+b}{2})^2}{(x-a)(x-b)}}{\frac{1}{x^2}} = \lim_{|x|\to\infty} \frac{2\log(x-\frac{a+b}{2}) - \log(x-a)(x-b)}{\frac{1}{x^2}}$$
$$= \lim_{|x|\to\infty} \frac{2\log(x-\frac{a+b}{2}) - \log(x^2-(a+b)x+ab)}{\frac{1}{x^2}}$$

We now use L'Hôpital:

$$= \lim_{|x|\to\infty} \frac{\frac{2}{x-\frac{a+b}{2}} - \frac{2x-a-b}{x^2-(a+b)x+ab}}{\frac{-2}{x^3}}$$
$$= \frac{-1}{2} \lim_{|x|\to\infty} \left(\frac{2x^3}{x-\frac{a+b}{2}} - \frac{2x^4-ax^3-bx^3}{x^2-(a+b)x+ab}\right)$$
$$= \frac{-1}{2} \lim_{|x|\to\infty} \left(\frac{2x^5-2(a+b)x^4+2abx^3}{x^3+P_{a,b}^2(x)} - \frac{2x^5-ax^4-bx^4-(a+b)x^4+(a+b)(\frac{a+b}{2})x^3}{x^3+P_{a,b}^2(x)}\right)$$
$$= \lim_{|x|\to\infty} \frac{-1}{2}\left(\frac{2abx^3-(\frac{a+b}{2})(a+b)x^3}{x^3+P_{a,b}^2(x)}\right) = \frac{1}{2}\frac{(\frac{a+b}{2})(a+b)-2ab}{1} = (\frac{a+b}{2})^2 - ab.$$

Which is constant, and hence $H(g)(x)$ is proportional to $|x|^{-2}$ as $|x| \to \infty$. □